Math 257 Definitions and Theorems

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1 Introduction to Linear Systems

Definition 1.1. A **linear equation** is a equation of the form

$$a_1x_1 + \ldots + a_nx_n = b$$

where $a_1, ..., a_n, b$ are numbers and $x_1, ..., x_n$ are variables.

Definition 1.2. A **linear system** is a collection of one or more linear equations involving the same set of variables, say, $x_1, x_2, ..., x_n$.

A **solution** of a linear system is a list $(s_1, s_2, ..., s_n)$ of numbers that makes each equation in the system true when the values $s_1, s_2, ..., s_n$ are substituted for $x_1, x_2, ..., x_n$, respectively.

Theorem 1.1. *A linear system has either*

***one unique solution, no solution** or **infinitely many solutions**.*

Definition 1.3. The **solution set** of a linear system is the set of all solutions of the linear system. Two linear systems are **equivalent** if they have the same solution set.

2 Matrices

Definition 2.1. An $m \times n$ **matrix** is a rectangular array of numbers with m rows and n columns.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \qquad a_{ij} \text{ is in the } i\text{th row and } j\text{th column of } A$$

Definition 2.2. Given a linear system,

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

we define its **coefficient** and **augmented** matrices to be

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \text{ and } \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right], \text{ respectively}$$

Definition 2.3. An **elementary row operation** is one of the following:

- (Replacement) Add a multiple of one row to another row: $R_i \to R_i + cR_j$, where $i \neq j$,
- (Interchange) Interchange two rows: $R_i \leftrightarrow R_j$,
- (Scaling) Multiply all entries in a row by a nonzero constant: $R_i \to cR_i$, where $c \neq 0$.

Remark. Every row operation is reversible.

Definition 2.4. Two matrices are said to be **row equivalent**, if one matrix can be transformed into the other matrix by a sequence of elementary row operations.

Theorem 2.1. *If the augmented matrices of two linear systems are row equivalent, then the two systems have the same solution set.*

3 Echelon Forms of Matrices

Definition 3.1. A matrix is in **echelon form (**or **row echelon form)** if

1. all *nonzero rows* (rows with at least one nonzero element) are above any rows of all zeros, and
2. the *leading entry* (the first nonzero number from the left) of a nonzero row is always strictly to the right of the leading entry of the row above it.

Definition 3.2. A matrix is in **row reduced echelon form** (or: **reduced echelon form**, or **RREF**) if it is in echelon form and the following as satisfied:

3. the leading entry in each nonzero row is 1, and

4. each leading entry is the only nonzero entry in its column.

Theorem 3.1 (Uniqueness of RREF)**.** *Each matrix is row-equivalent to one and only one matrix in reduced echelon form.*

Definition 3.3. We say a matrix B is **the reduced echelon form** (or: the **RREF**) of a matrix A if A and B are row-equivalent and B is in reduced echelon form.

Definition 3.4. A **pivot position** is the position of a leading entry in an echelon form of a matrix. A **pivot column** is a column that contains a pivot position.

Definition 3.5. A **basic variable** (or **pivot variable**) is a variable that corresponds to a pivot column in the coefficient matrix of a linear system. A **free variable** is variable that is *not* a pivot variable.

4 Gaussian Elimination

Goal. Solve linear systems for the pivot variables in terms of the free variables (if any) in the equation.

Algorithm 4.1. (Gaussian Elimination) Given a linear system,

(1) write down the augmented matrix,

(2) find the reduced echelon form of the matrix,

(3) write down the equations corresponding to the reduced echelon form, and

(4) express pivot variables in terms of free variables.

Theorem 4.1. *A linear system is **consistent** if and only if an echelon form of the augmented matrix has **no** row of the form* $\left[\begin{array}{ccc|c} 0 & \ldots & 0 & b \end{array}\right]$*, where b is nonzero. If a linear system is consistent, then the linear system has*

- *a unique solution (when there are **no** free variables), or*
- *infinitely many solutions (when there is **at least one** free variable).*

5 Linear Combinations

Definition 5.1. Consider $m \times n$-matrices

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \text{ and } B = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{bmatrix}.$$

- We define their **sum** $A + B$ to be

$$A + B := \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \cdots & a_{2n} + b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \cdots & a_{mn} + b_{mn} \end{bmatrix}.$$

- If $c \in \mathbb{R}$ is a scalar, we define the ***scalar* product** cA to be

$$cA := \begin{bmatrix} ca_{11} & ca_{12} & \cdots & ca_{1n} \\ ca_{21} & ca_{22} & \cdots & ca_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ ca_{m1} & ca_{m2} & \cdots & ca_{mn} \end{bmatrix}.$$

Warning. Addition is only defined if A has the same number of columns and rows as B.

Definition 5.2. A **column vector** is an $m \times 1$-matrix. A **row vector** is a $1 \times n$-matrix.

Definition 5.3. If A is $m \times n$, the **transpose** of A is the $n \times m$ matrix, denoted by A^T, whose columns are formed from the corresponding rows of A. In terms of matrix elements: $(A^T)_{ij} = A_{ji}$.

Remark. The transpose of a column vector is a row vector and vice versa.

Definition 5.4. The **linear combination** of $m \times n$-matrices $A_1, A_2, \ldots, A_p$ with **coefficients** $c_1, c_2, \ldots, c_p$ is defined as:

$$c_1 A_1 + c_2 A_2 + \cdots + c_p A_p.$$

Definition 5.5. Given $m \times n$ matrices $A_1, \ldots, A_p$, we define their **Span**, $\text{Span}\,(A_1, \ldots, A_p)$, to be the set of all linear combinations of $A_1, \ldots, A_p$. That is,

$$\text{Span}\,(A_1, \ldots, A_p) \ := \{c_1 A_1 + c_2 A_2 + \cdots + c_p A_p \ : \ c_1, \ldots, c_p \text{ scalars}\}.$$

Definition 5.6. We denote the set of all column vectors of length m by $\mathbb{R}^m$.

Remark. Solving linear systems is the same as finding linear combinations!

Theorem 5.1. *Let $a_1, \ldots, a_n, b \in \mathbb{R}^m$. A vector equation*

$$x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \cdots + x_n \mathbf{a}_n = \mathbf{b}$$

has the same solution set as the linear system whose augmented matrix is

$$\left[\begin{array}{cccc|c} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n & \mathbf{b} \end{array} \right]$$

In particular, $\mathbf{b}$ can be generated by a linear combination of $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$ if and only if there is a solution to the linear system corresponding to the augmented matrix.

6 Matrix-vector multiplication

Definition 6.1. Let $\mathbf{x}$ be a vector in $\mathbb{R}^n$ and $A = \begin{bmatrix}\mathbf{a}_1 & \dots & \mathbf{a}_n\end{bmatrix}$ an $m \times n$-matrix. We define the product $A\mathbf{x}$ by

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \ldots + x_n\mathbf{a}_n.$$

Remark.

- $A\mathbf{x}$ is a linear combination of the columns of A using the entries in $\mathbf{x}$ as coefficients.
- $A\mathbf{x}$ is only defined if the number of entries of $\mathbf{x}$ is equal to the number of columns of A.

Theorem 6.1. *Let $A = \begin{bmatrix}\mathbf{a}_1 & \dots & \mathbf{a}_n\end{bmatrix}$ be an $m \times n$-matrix and $\mathbf{b}$ in $\mathbb{R}^m$. Then the following are equivalent:*

- *$(x_1, x_2, \ldots, x_n)$ is a solution of the vector equation $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$*
- $\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ *is a solution to the matrix equation $A\mathbf{x} = \mathbf{b}$.*
- *$(x_1, x_2, \ldots, x_n)$ is a solution of the system with augmented matrix $\left[\begin{array}{c|c} A & \mathbf{b} \end{array}\right]$*

Notation. We will write $A\mathbf{x} = \mathbf{b}$ for the system of equations with augmented matrix $\left[\begin{array}{c|c} A & \mathbf{b} \end{array}\right]$.

7 Matrix Multiplication

Definition 7.1. Let A be an $m \times n$-matrix and let $B = \begin{bmatrix}\mathbf{b}_1 & \dots & \mathbf{b}_p\end{bmatrix}$ be an $n \times p$-matrix. We define

$$AB := [A\mathbf{b}_1 \;\; A\mathbf{b}_2 \;\; \cdots \;\; A\mathbf{b}_p]$$

Theorem 7.1. *Let A be an $m \times n$ matrix and B be an $n \times p$ matrix. Then for every $\mathbf{x} \in \mathbb{R}^p$*

$$A(B\mathbf{x}) = (AB)\mathbf{x}.$$

Note. Other ways to compute AB.

- **Row-Column Rule** Let A be $m \times n$ and B be $n \times p$ such that

$$A = \begin{bmatrix} \mathbf{R}_1 \\ \vdots \\ \mathbf{R}_m \end{bmatrix}, \text{ and } B = \begin{bmatrix}\mathbf{C}_1 & \dots & \mathbf{C}_p\end{bmatrix}. \text{ Then}$$

$$AB = \begin{bmatrix} \mathbf{R}_1\mathbf{C}_1 & \dots & \mathbf{R}_1\mathbf{C}_p \\ \mathbf{R}_2\mathbf{C}_1 & \dots & \mathbf{R}_2\mathbf{C}_p \\ \mathbf{R}_m\mathbf{C}_1 & \dots & \mathbf{R}_m\mathbf{C}_p \end{bmatrix}$$

and $(AB)_{ij} = \mathbf{R}_i\mathbf{C}_j = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$.

- **Outer Product Rule** Let A be $m \times n$ and B be $n \times p$ such that

$$A = \begin{bmatrix} \mathbf{C}_1 & \cdots & \mathbf{C}_n \end{bmatrix}, \text{ and } B = \begin{bmatrix} \mathbf{R}_1 \\ \vdots \\ \mathbf{R}_n \end{bmatrix}. \text{ Then}$$

$$AB = \mathbf{C}_1\mathbf{R}_1 + \cdots + \cdots + \mathbf{C}_n\mathbf{R}_n$$

8 Properties of Matrix Multiplication

Definition 8.1. The identity matrix I_n of size n is defined as

$$I_n = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}.$$

Theorem 8.1. *Let A be an $m \times n$ matrix and let B and C be matrices for which the indicated sums and products are defined.*

(a) $A(BC) = (AB)C$ *(associative law of multiplication)*

(b) $A(B + C) = AB + AC$, $(B + C)A = BA + CA$ *(distributive laws)*

(c) $r(AB) = (rA)B = A(rB)$ *for every scalar r,*

(d) $A(rB + sC) = rAB + sAC$ *for every scalars r, s (linearity of matrix multiplication)*

(e) $I_m A = A = AI_n$ *(identity for matrix multiplication)*

Warning. Properties above are analogous to properties of real numbers. But **NOT ALL** properties of real number also hold for matrices. In particular, AB need not equal BA.

Theorem 8.2. *The transpose of a product is the product of transposes **in opposite order**:*

$$(AB)^T = B^T A^T$$

Definition 8.2. Let A be a square matrix. We write A^k for $A \cdots A$, k-times; that is A^k is obtained by multiplying A k-times with itself.

9 Elementary Matrices

Definition 9.1. An **elementary matrix** is one that is obtained by performing a single elementary row operation on an identity matrix.
A **permutation matrix** is one that is obtained by performing row exchanges on an identity matrix.

Theorem 9.1. *If an elementary row operation is performed on an $m \times n$-matrix A, the resulting matrix can be written as EA, where the $m \times m$-matrix E is created by performing the same row operations on I_m.*

Theorem 9.2. *Let A, B be two $m \times n$-matrices and row-equivalent. Then there is a sequence $m \times m$-elementary matrices $E_1, \ldots, E_\ell$ such that*

$$E_\ell \ldots E_1 A = B.$$

10 Inverse of a Matrix

Remark. The *inverse* of a real number a is denoted by a^{-1}. For example, $7^{-1} = 1/7$ and $7 \cdot 7^{-1} = 7^{-1} \cdot 7 = 1$. Note that not all real numbers have inverse. Namely, 0^{-1} is not defined as there is no real number b such that $0 \cdot b = 1$.

Definition 10.1. An $n \times n$ matrix A is said to be **invertible** if there is an $n \times n$ matrix C satisfying

$$CA = AC = I_n$$

where I_n is the $n \times n$ identity matrix. We call C the **inverse** of A.

Theorem 10.1. *Let A be an invertible matrix, then its inverse C is unique.*

Theorem 10.2. *Suppose A and B are invertible. Then:*

(a) A^{-1} is invertible and $\left(A^{-1}\right)^{-1} = A$ (i.e. A is the inverse of A^{-1}).

(b) AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.

(c) A^T is invertible and $\left(A^T\right)^{-1} = \left(A^{-1}\right)^T$.

Theorem 10.3. *Let A be an invertible $n \times n$ matrix. Then for each $\mathbf{b}$ in $\mathbb{R}^n$, the equation $A\mathbf{x} = \mathbf{b}$ has the unique solution $\mathbf{x} = A^{-1}\mathbf{b}$.*

11 Computing the Inverse

Theorem 11.1. *Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $ad - bc \neq 0$, then A is invertible and*

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

If $ad - bc = 0$, then A is not invertible.

Theorem 11.2. *Let A be an $n \times n$-matrix. The following are equivalent:*

- *A is invertible.*
- *the reduced echelon form of A is I_n.*

Theorem 11.3. *Suppose A is invertible. The every sequence of elementary row operations that reduces A to I_n will also transform I_n to A^{-1}.*

Algorithm 11.1.

- Place A and I side-by-side to form an augmented matrix $[\ A \mid I\ ]$. This is an $n \times 2n$ matrix (Big Augmented Matrix), instead of $n \times (n+1)$.
- Perform row operations on this matrix (which will produce identical operations on A and I).
- By Theorem:

  $$[\ A \mid I\ ] \text{ will row reduce to } \left[\ I \mid A^{-1}\ \right]$$

  or A is not invertible.

12 LU Decomposition

Definition 12.1. An $n \times n$ matrix A is called

- **upper triangular** if it is of the form

$$\begin{bmatrix} \star & \star & \star & \star & \star \\ 0 & \star & \star & \star & \star \\ 0 & 0 & \star & \star & \star \\ 0 & 0 & 0 & \ddots & \vdots \\ 0 & 0 & 0 & 0 & * \end{bmatrix},$$

- **lower triangular** if it is of the form

$$\begin{bmatrix} \star & 0 & 0 & 0 & 0 \\ \star & \star & 0 & 0 & 0 \\ \star & \star & \star & 0 & 0 \\ \star & \star & \star & \ddots & \vdots \\ \star & \star & \star & \star & \star \end{bmatrix}.$$

Theorem 12.1. *The product of two lower (upper) triangular matrices is lower (upper) triangular.*

Remark. The inverse of a lower (upper) triangular matrix (if it exists) is again lower (upper) triangular.

Definition 12.2. A matrix A has an **LU decomposition** if there is a lower triangular matrix L and a upper triangular matrix U such that $A = LU$.

Theorem 12.2. *Let A be an $n \times n$-matrix. If A can be brought to echelon form just using row operations of the form $R_i \to R_i + cR_j$ where $j < i$, then A has an LU-decomposition.*

Remark. Not every matrix has an LU decomposition. The LU decomposition of a matrix is not unique.

13 Solving Linear Systems using LU Decomposition

Theorem 13.1. *Let A be an $n \times n$-matrix such that $A = LU$, where L is a lower triangular matrix and U is a upper triangular matrix, and let $\mathbf{b} \in \mathbb{R}^n$. In order to find a solution of the linear system*

$$A\mathbf{x} = \mathbf{b},$$

it is enough to find a solution of the linear system

$$U\mathbf{x} = \mathbf{c},$$

where $\mathbf{c}$ satisfies $L\mathbf{c} = \mathbf{b}$.

Theorem 13.2. *Let A be $n \times n$ matrix. Then there is a permutation matrix P such that PA has an LU-decomposition.*

Spring-Mass Systems

See the notes for information about this application.

14 Inner Product and Orthogonality

Definition 14.1. The **inner product** of $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ is

$$\mathbf{v} \cdot \mathbf{w} = \mathbf{v}^T\mathbf{w}.$$

Theorem 14.1. *Let $\mathbf{u}, \mathbf{v}$ and $\mathbf{w}$ be vectors in $\mathbb{R}^n$, and let c be any scalar. Then*

(a) $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$

(b) $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$

(c) $(c\mathbf{u}) \cdot \mathbf{v} = c\,(\mathbf{u} \cdot \mathbf{v}) = \mathbf{u} \cdot (c\mathbf{v})$

(d) $\mathbf{u} \cdot \mathbf{u} \geq \mathbf{0}$, *and* $\mathbf{u} \cdot \mathbf{u} = \mathbf{0}$ *if and only if* $\mathbf{u} = \mathbf{0}$.

Definition 14.2. Let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$.

- The **norm** (or **length**) of $\mathbf{v}$ is

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + \cdots + v_n^2}.$$

- The **distance** between $\mathbf{v}$ and $\mathbf{w}$ is

$$\operatorname{dist}(\mathbf{v}, \mathbf{w}) = \|\mathbf{v} - \mathbf{w}\|.$$

Definition 14.3. Let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$. We say $\mathbf{v}$ and $\mathbf{w}$ are **orthogonal** if $\mathbf{v} \cdot \mathbf{w} = 0$.

Theorem 14.2. *Let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ be non-zero. Then $\mathbf{v}$ and $\mathbf{w}$ are orthogonal if and only if they are perpendicular (ie. if they form a right angle).*

Definition 14.4. A set of vectors in $\mathbb{R}^n$ is **pairwise orthogonal** if each pairing of them is orthogonal. We call such a set an **orthogonal set**.

Definition 14.5. A **unit vector** in $\mathbb{R}^n$ is vector of length 1.

Definition 14.6. A set of vectors form an **orthonormal set** if this set is an orthogonal set and all vectors in the set are unit vectors.

15 Subspaces

Definition 15.1. A non-empty subset H of $\mathbb{R}^n$ is a **subspace** of $\mathbb{R}^n$ if it satisfies the following two conditions:

- if $\mathbf{u}, \mathbf{v} \in H$, then the sum $\mathbf{u} + \mathbf{v} \in H$. ($H$ is *closed under vector addition*), and
- if $\mathbf{u} \in H$ and c is a scalar, then $c\mathbf{u} \in H$. (H is *closed under scalar multiplication.*)

Theorem 15.1. *Let $\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_m} \in \mathbb{R}^n$. Then Span $(\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_m})$ is a subspace of $\mathbb{R}^n$.*

16 Column Spaces and Null Spaces

Definition 16.1. The **column space**, written as $\mathrm{Col}(A)$, of an $m \times n$ matrix A is the set of all linear combinations of the columns of A. If $A = \begin{bmatrix} \mathbf{a_1} & \mathbf{a_2} & \cdots & \mathbf{a_n} \end{bmatrix}$, then $\mathrm{Col}(A) = \mathrm{Span}\,(\mathbf{a_1}, \mathbf{a_2}, \ldots, \mathbf{a_n})$.

Theorem 16.1. *Let A be an $m \times n$ matrix. Then $\mathrm{Col}(A)$ is a subspace of $\mathbb{R}^m$.*

Theorem 16.2. *Let A be an $m \times n$ matrix and $\mathbf{b} \in \mathbb{R}^m$. Then $\mathbf{b}$ is in $\mathrm{Col}(A)$ if and only if the linear system $A\mathbf{x} = \mathbf{b}$ has a solution.*

Definition 16.2. The **nullspace** of an $m \times n$ matrix A, written as $\mathrm{Nul}(A)$, is the set of all solutions to the homogeneous equation $A\mathbf{x} = \mathbf{0}$; that is, $\mathrm{Nul}(A) = \{\mathbf{v} \in \mathbb{R}^n \ : \ A\mathbf{v} = \mathbf{0}\}$.

Theorem 16.3. *The nullspace of an $m \times n$ matrix A is a subspace of $\mathbb{R}^n$.*

Theorem 16.4. *Let A be an $m \times n$ matrix, let $\mathbf{b} \in \mathbb{R}^m$, and let $\mathbf{w} \in \mathbb{R}^n$ such that $A\mathbf{w} = \mathbf{b}$. Then $\{\mathbf{v} \in \mathbb{R}^n \ : \ A\mathbf{v} = \mathbf{b}\} = \mathbf{w} + \mathrm{Nul}(A)$.*

17 Abstract Vector Spaces

Definition 17.1. A **vector space** is a non-empty set V of objects, called **vectors**, for which linear combinations make sense. More precisely, there are two operations on V, called *addition* and *multiplication* by scalars (real numbers), satisfying the following axioms for all $u, v, w \in V$ and for all scalars $c, d \in \mathbb{R}$:

- $\mathbf{u} + \mathbf{v}$ is in V. (V is "closed under addition".)
- $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.
- $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$.
- There is a vector (called the zero vector) $\mathbf{0}_V$ in V such that $\mathbf{u} + \mathbf{0}_V = \mathbf{u}$.
- For each $\mathbf{u}$ in V, there is a vector $-\mathbf{u}$ in V satisfying $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}_V$.
- $c\mathbf{u}$ is in V. (V is "closed under scalar multiplication".)
- $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$.
- $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$.
- $(cd)\mathbf{u} = c(d\mathbf{u})$.
- $1\mathbf{u} = \mathbf{u}$.

Definition 17.2. Let V be a vector space. A non-empty subset $W \subseteq V$ is a **subspace** of V if

- $\mathbf{u} + \mathbf{v}$ for all $\mathbf{u}, \mathbf{v} \in U$ (*closed under addition*)
- $c\mathbf{u}$ for all $\mathbf{u} \in U$ and $c \in \mathbb{R}$ (*closed under scalar multiplication*)

18 Linear Independence

Definition 18.1. Vectors $\mathbf{v}_1, \ldots, \mathbf{v}_p$ are said to be **linearly independent** if the equation

$$x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \cdots + x_p\mathbf{v}_p = \mathbf{0}$$

has only the trivial solution (namely, $x_1 = x_2 = \cdots = x_p = 0$). We say the vectors are **linearly dependent** if they are not linearly independent.

Theorem 18.1. *Vectors $\mathbf{v}_1, \ldots, \mathbf{v}_p$ are linearly dependent if and only if there is $i \in \{1, \ldots, p\}$ such that $\mathbf{v}_i \in \mathrm{Span}(\mathbf{v}_1, \ldots, \mathbf{v}_{i-1}, \mathbf{v}_{i+1}, \ldots, \mathbf{v}_p)$.*

Theorem 18.2. *Let A be an $m \times n$ matrix. The following are equivalent:*

- *The columns of A are linearly independent.*
- *$A\mathbf{x} = \mathbf{0}$ has only the solution $\mathbf{x} = \mathbf{0}$.*
- *A has n pivots.*
- *there are no free variables for $A\mathbf{x} = \mathbf{0}$.*

19 Bases and Dimension

Definition 19.1. Let V be a vector space. A sequence of vectors $(\mathbf{v}_1, \ldots, \mathbf{v}_p)$ in V is a **basis** of V if

- $V = \text{Span}(\mathbf{v}_1, \ldots, \mathbf{v}_p)$, and
- $(\mathbf{v}_1, \ldots, \mathbf{v}_p)$ are linearly independent.

Theorem 19.1. *Every two bases in a vector space V contain the same numbers of vectors.*

Definition 19.2. The number of vectors in a basis of V is the **dimension** of V.

Theorem 19.2. *Suppose that V has dimension d. Then*

- *A sequence of d vectors in V are a basis if they span V.*
- *A sequence of d vectors in V are a basis if they are linearly independent.*

Theorem 19.3. *A basis is a **minimal** spanning set of V; that is the elements of the basis span V but you cannot delete any of these elements and still get all of V.*

20 The Four Fundamental Subspaces

Algorithm 20.1. To find a basis for $\text{Nul}(A)$:

- Find the parametric form of the solutions to $A\mathbf{x} = \mathbf{0}$.
- Express solutions $\mathbf{x}$ as a linear combination of vectors with the free variables as coefficients.
- Use these vectors as a basis of $\text{Nul}(A)$.

Definition 20.1. The **rank** of a matrix is the number of pivots it has.

Theorem 20.1. *Let A be an $m \times n$ matrix with rank r. Then* $\dim \text{Nul}(A) = n - r$.

Theorem 20.2. *Let A be an $m \times n$ matrix with rank r. The pivot columns of A form a basis of* $\text{Col}(A)$. *In particular,* $\dim \text{Col}(A) = r$.

Theorem 20.3. *Let A, B be two row-equivalent matrices. Then* $\text{Col}(A^T) = \text{Col}(B^T)$.

Theorem 20.4. *Let A be $m \times n$ matrix with rank r. Then the non-zero rows in an echelon form of A form a basis of* $\text{Col}(A^T)$, *and thus* $\dim(A^T) = r$.

Theorem 20.5. *Let A be an $m \times n$ matrix with rank r. Then*

- $\dim \text{Col}(A) = \dim \text{Col}(A^T) = r$, *and*
- $\dim \text{Nul}(A) = n - r$.

Graphs

Definition 20.2. A **graph** is a set of nodes (or: vertices) that are connected through edges.

Definition 20.3. Let $\mathcal{G}$ be a graph with n nodes. The **adjacency matrix** of $\mathcal{G}$ is the $n \times n$-matrix $A = (a_{ij})$ such that

$$a_{ij} = \begin{cases} 1 & \text{if there is an edge between node } i \text{ and node } j \\ 0 & \text{otherwise .} \end{cases}$$

Definition 20.4. A **walk of length** k on a graph of is a sequence of $k+1$ vertices and k edges between two nodes (including the start and end) that may repeat. A **path** is walk in which all vertices are distinct.

Definition 20.5. A graph is **connected** if for every pair of nodes i and j there is a walk from node i to node j. A graph is **disconnected** if it is not connected.

Theorem 20.6. *Let $\mathcal{G}$ be a graph and let A be its adjacency matrix. Then the entry in the i-th row and j-th column of A^ℓ is the number of walks of length ℓ from node j to node i on $\mathcal{G}$.*

Definition 20.6. A **directed graph** is a set of vertices connected by edges, where the edges have a direction associated with them.

Definition 20.7. Let G be a directed graph with m edges and n nodes. The **adjacency matrix** of G is the $n \times n$ matrix $A = (a_{i,j})_{i,j}$ with

$$a_{i,j} = \begin{cases} 1, & \text{if there is a directed edge from node } j \text{ to node } i \\ 0, & \text{otherwise} \end{cases}$$

Definition 20.8. Let G be a directed graph with m edges and n nodes. The **edge-node incidence matrix** of G is the $m \times n$ matrix $A = (a_{i,j})_{i,j}$ with

$$a_{i,j} = \begin{cases} -1, & \text{if edge } i \text{ leaves node } j \\ +1, & \text{if edge } i \text{ enters node } j \\ 0, & \text{otherwise} \end{cases}$$

Definition 20.9. A **connected component** of an undirected graph is a part in which any two vertices are connected to each other by paths, and which is connected to no additional vertices in the rest of the graph. The connected components of a directed graph are those of its underlying undirected graph. A graph is **connected** if only has one connected component.

Theorem 20.7. *Let $\mathcal{G}$ be a directed graph and let A be its edge-node incidence matrix. Then* $\dim \mathrm{Nul}(A)$ *is equal to the number of connected components of $\mathcal{G}$.*

Definition 20.10. A **cycle** in an undirected graph is a path in which all edges are distinct and the only repeated vertices are the first and last vertices. By cycles of a directed graph we mean those of its underlying undirected graph.

Definition 20.11. The span of all cycle vectors of a graph $\mathcal{G}$ is called the **cycle space**.

Theorem 20.8. *Let $\mathcal{G}$ be a directed graph and let A be its edge-node incidence matrix. Then the cycle space of $\mathcal{G}$ is equal to* $\text{Nul}(A^T)$.

21 Orthogonal Complements

Definition 21.1. Let W be a subspace of $\mathbb{R}^n$. The **orthogonal complement** of W is the subspace $W^\perp$ of all vectors that are orthogonal to W; that is

$$W^\perp := \{\mathbf{v} \in \mathbb{R}^n \ : \ \mathbf{v} \cdot \mathbf{w} = 0 \text{ for all } \mathbf{w} \in W\}.$$

Remark. Observe that $(W^\perp)^\perp = W$.

Theorem 21.1. *Let A be an $m \times n$ matrix. Then* $\text{Nul}(A)$ *is the orthogonal complement of* $\text{Col}(A^T)$*; that is* $\text{Nul}(A) = \text{Col}(A^T)^\perp$.

Remark. It follows that

- $\text{Nul}(A)^\perp = \text{Col}(A^T)$.
- $\text{Nul}(A^T) = \text{Col}(A)^\perp$.

Theorem 21.2. *Let V be a subspace of $\mathbb{R}^n$. Then* $\dim V + \dim V^\perp = n$.

22 Coordinates

Theorem 22.1. *Let $(\mathbf{v}_1, \ldots, \mathbf{v}_p)$ be a basis of V. Then every vector $\mathbf{w}$ in V can be expressed **uniquely** as*

$$\mathbf{w} = c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p.$$

Definition 22.1. Let $\mathcal{B} = (\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p)$ be an (ordered) basis of V, and let $\mathbf{w} \in V$. The **coordinate vector $\mathbf{w}_\mathcal{B}$** of $\mathbf{w}$ with respect to the basis $\mathcal{B}$ is

$$\mathbf{w}_\mathcal{B} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \\ c_p \end{bmatrix}, \quad \text{if } \mathbf{w} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_p\mathbf{v}_p.$$

Definition 22.2. In $\mathbb{R}^n$ let $\mathbf{e}_i$ denote the vector with a 1 in the i-th coordinate and 0's elsewhere. The **standard basis of** $\mathbb{R}^n$ is the ordered basis $\mathcal{E}_n := (\mathbf{e}_1, \ldots, \mathbf{e}_n)$.

Definition 22.3. Let $\mathcal{B}$ and $\mathcal{C}$ be two bases of $\mathbb{R}^n$. The **change of basis matrix** $I_{\mathcal{C},\mathcal{B}}$ is the matrix such that for all $\mathbf{v} \in \mathbb{R}^n$

$$I_{\mathcal{C},\mathcal{B}}\mathbf{v}_\mathcal{B} = \mathbf{v}_\mathcal{C}$$

Theorem 22.2. *Let $\mathcal{B} = (\mathbf{b}_1, \ldots, \mathbf{b_n})$ be a basis of $\mathbb{R}^n$. Then*

$$I_{\mathcal{E}_n,\mathcal{B}} = \begin{bmatrix} \mathbf{b}_1 & \ldots & \mathbf{b}_n \end{bmatrix}$$

That is, for all $\mathbf{v} \in \mathbb{R}^n$,

$$\mathbf{v} = \begin{bmatrix} \mathbf{b}_1 & \ldots & \mathbf{b}_n \end{bmatrix} \mathbf{v}_{\mathcal{B}}.$$

Remark. Let $\mathcal{B}$ and $\mathcal{C}$ be bases in $\mathbb{R}^n$.

- To compute $I_{\mathcal{B},\mathcal{E}_n}$, notice that

$$\mathbf{v} = I_{\mathcal{E}_n,\mathcal{B}}\mathbf{v}_{\mathcal{B}} \rightsquigarrow I^{-1}_{\mathcal{E}_n,\mathcal{B}}\mathbf{v} = \mathbf{v}_{\mathcal{B}} \rightsquigarrow I_{\mathcal{B},\mathcal{E}_n} = I^{-1}_{\mathcal{E}_n,\mathcal{B}}$$

- To compute $I_{\mathcal{B},\mathcal{C}}$, we can either use

$$I_{\mathcal{B},\mathcal{E}_n} I_{\mathcal{E}_n,\mathcal{C}}\mathbf{v}_{\mathcal{C}} = I_{\mathcal{B},\mathcal{E}_n}\mathbf{v} = \mathbf{v}_{\mathcal{B}} \rightsquigarrow I_{\mathcal{B},\mathcal{C}} = I_{\mathcal{B},\mathcal{E}_n} I_{\mathcal{E}_n,\mathcal{C}}$$

  or use

$$I_{\mathcal{C},\mathcal{B}} = \begin{bmatrix} (\mathbf{b}_1)_{\mathcal{C}} & \ldots & (\mathbf{b}_n)_{\mathcal{C}} \end{bmatrix}$$

23 Orthogonal and Orthonormal Bases

Theorem 23.1. *Let $\mathbf{v}_1, \ldots, \mathbf{v}_m \in \mathbb{R}^n$ be non-zero and pairwise orthogonal. Then $\mathbf{v}_1, \ldots, \mathbf{v}_m$ are linearly independent.*

Remark. The theorem implies that a set of n orthonormal vectors in $\mathbb{R}^n$ is a basis of $\mathbb{R}^n$.

Definition 23.1. An **orthogonal basis** (an **orthonormal basis**) is an orthogonal set of vectors (an orthonormal set of vectors) that forms a basis.

Remark. When $\mathcal{B}$ is orthonormal, then $\mathbf{b}_i \cdot \mathbf{b}_i = 1$ for $i = 1, \ldots, n$.

Theorem 23.2. *Let $\mathcal{B} := (\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n)$ be an orthogonal basis of $\mathbb{R}^n$, and let $\mathbf{v} \in \mathbb{R}^n$. Then*

$$\mathbf{v} = \frac{\mathbf{v} \cdot \mathbf{b}_1}{\mathbf{b}_1 \cdot \mathbf{b}_1}\mathbf{b}_1 + \ldots + \frac{\mathbf{v} \cdot \mathbf{b}_n}{\mathbf{b}_n \cdot \mathbf{b}_n}\mathbf{b}_n.$$

Theorem 23.3. *Let $\mathcal{U} = (\mathbf{u}_1, \ldots, \mathbf{u_n})$ be an orthonormal basis of $\mathbb{R}^n$. Then*

$$I_{\mathcal{U},\mathcal{E}_n} = \begin{bmatrix} \mathbf{u}_1 & \ldots & \mathbf{u}_n \end{bmatrix}^T.$$

Definition 23.2. An $n \times n$-matrix Q is **orthogonal** if $Q^{-1} = Q^T$.

24 Linear Transformations

Definition 24.1. Let V and W be vector spaces. A map $T : V \to W$ is a **linear transformation** if

$$T(a\mathbf{v} + b\mathbf{w}) = aT(\mathbf{v}) + bT(\mathbf{w})$$

for all $\mathbf{v}, \mathbf{w} \in V$ and all $a, b \in \mathbb{R}$.

Remark. $T(\mathbf{0}_V) = T(0 \cdot \mathbf{0}_V) = 0 \cdot T(\mathbf{0}_V) = \mathbf{0}_W \rightsquigarrow T(\mathbf{0}_V) = \mathbf{0}_W$

Theorem 24.1. *Let V, W be two vector spaces, let $T : V \to W$ be a linear transformation and let $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be a basis of V. Then T is completely determined by the values $T(\mathbf{v}_1), \ldots, T(\mathbf{v}_n)$.*

Theorem 24.2. *Let $T\colon \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. Then there is a $m \times n$ matrix A such that*

- $T(\mathbf{v}) = A\mathbf{v}$, *for all* $\mathbf{v} \in \mathbb{R}^n$.
- $A = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \ldots & T(\mathbf{e}_n) \end{bmatrix}$, *where $(\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n)$ is the standard basis of $\mathbb{R}^n$.*

Remark. We call this A the **coordinate matrix of** T with respect to the standard bases - we write $T_{\mathcal{E}_m, \mathcal{E}_n}$.

25 Coordinate Matrices of Linear Transformations

Theorem 25.1. *Let V, W be two vector space, let $\mathcal{B} = (\mathbf{b}_1, \ldots, \mathbf{b}_n)$ be a basis of V and $\mathcal{C} = (\mathbf{c}_1, \ldots, \mathbf{c}_m)$ be a basis of W, and let $T\colon V \to W$ be a linear transformation. Then there is a $m \times n$ matrix $T_{\mathcal{C},\mathcal{B}}$ such that*

- $T(\mathbf{v})_{\mathcal{C}} = T_{\mathcal{C},\mathcal{B}}\mathbf{v}_{\mathcal{B}}$, *for all* $\mathbf{v} \in V$.
- $T_{\mathcal{C},\mathcal{B}} = \begin{bmatrix} T(\mathbf{b}_1)_{\mathcal{C}} & T(\mathbf{b}_2)_{\mathcal{C}} & \ldots & T(\mathbf{b}_n)_{\mathcal{C}} \end{bmatrix}$.

Theorem 25.2. *Let $T : \mathbb{R}^m \to \mathbb{R}^n$ be a linear transformation and $\mathcal{A}$ and $\mathcal{B}$ be two bases of $\mathbb{R}^m$ and $\mathcal{C}, \mathcal{D}$ be two bases of $\mathbb{R}^n$. Then*

$$T_{\mathcal{C},\mathcal{A}} = I_{\mathcal{C},\mathcal{D}} T_{\mathcal{D},\mathcal{B}} I_{\mathcal{B},\mathcal{A}}.$$

26 Determinants

Goal: Define a determinant function that takes an $n \times n$ matrix A to a real number $\det(A)$, such that $\det(A) \neq 0$ if and only if A is invertible.

We start with small matrices:

Definition 26.1. The **determinant** of

- a 1×1 matrix is $\det([a]) = a$.
- a 2×2 matrix is $\det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = ad - bc$.

Notation. We will write both $\det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right)$ and $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ for the determinant.

Definition 26.2. The **determinant** is the function that assigns to each $n \times n$-matrix a number and satisfies the following conditions:

- **(Normalization)** $\det I_n = 1$,
- It is affected by elementary row operations as follows:
  - **(Replacement)** Adding a multiple of one row to another row *does not change* the determinant.
  - **(Interchange)** Interchanging two different rows *reverses* the sign of the determinant.
  - **(Scaling)** Multiplying all entries in a row by s, *multiplies* the determinant by s.

Theorem 26.1. *The determinant of a triangular matrix is the product of the diagonal entries.*

Theorem 26.2. *Let A be an $n \times n$-matrix, then $\det(A) = 0$ if and only if A is not invertible.*

Theorem 26.3. *Let A, B be two $n \times n$-matrices. Then $\det(AB) = \det(A)\det(B)$.*

Theorem 26.4. *Let A be an $n \times n$-matrix. Then $\det(A^T) = \det(A)$.*

27 Cofactor expansion

Notation. Let A be an $n \times n$-matrix. We denote by A_{ij} the matrix obtained from matrix A by deleting the i-th row and j-th column of A.

Definition 27.1. Let A be an $n \times n$-matrix. The **$(\mathbf{i}, \mathbf{j})$-cofactor** of A is the scalar C_{ij} defined by

$$C_{ij} = (-1)^{i+j} \det A_{ij}.$$

Theorem 27.1. *Let A be an $n \times n$-matrix. Then for every $i, j \in \{1, \ldots, n\}$*

$$\begin{aligned} \det A &= a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in} \qquad (\textit{expansion across row } i) \\ &= a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj} \qquad (\textit{expansion down column } j) \end{aligned}$$

28 Eigenvectors

Definition 28.1. Let A be an $n \times n$ matrix. An **eigenvector** of A is a *nonzero* $\mathbf{v} \in \mathbb{R}^n$ such that

$$A\mathbf{v} = \lambda\mathbf{v},$$

where λ is a scalar, known as the **eigenvalue** associated with $\mathbf{v}$.

Definition 28.2. Let λ be an eigenvalue of A. The **eigenspace** of A associated with λ is the set

$$\mathrm{Eig}_\lambda(A) := \{\mathbf{v} \ : \ A\mathbf{v} = \lambda\mathbf{v}\}.$$

It consists of all the eigenvectors of A with eigenvalue λ and the zero vector.

29 Computing Eigenvalues and Eigenvectors

Theorem 29.1. *Let A be an $n \times n$ matrix and λ be a scalar. Then λ is an eigenvalue of A if and only if* $\det(A - \lambda I) = 0$.

Theorem 29.2. *Let A be an $n \times n$ matrix. Then $p_A(t) := \det(A - tI)$ is a polynomial of degree n. Thus A has at most n eigenvalues.*

Definition 29.1. We call $p_A(t)$ the **characteristic polynomial** of A.

Theorem 29.3. *Let A be $n \times n$ matrix and let λ be eigenvalue of A. Then*

$$Eig_\lambda(A) = Nul(A - \lambda I).$$

30 Properties of Eigenvectors and Eigenvalues

Definition 30.1. Let A be an $n \times n$ matrix and let λ be an eigenvalue of A.

- The **algebraic multiplicity** of λ is its multiplicity as a root of the characteristic polynomial, that is, the largest integer k such that $(t-\lambda)^k$ divides $p_A(t)$.
- The **geometric multiplicity** of λ is the dimension of the eigenspace $\mathrm{Eig}_\lambda(A)$ of λ.

Theorem 30.1. *Let A be an $n \times n$ matrix and let $\mathbf{v}_1, \ldots, \mathbf{v}_m$ be eigenvectors of A corresponding to different eigenvalues. Then $\mathbf{v}_1, \ldots, \mathbf{v}_m$ are linearly independent.*

Definition 30.2. Let

$$A = \begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \ldots & a_{nn} \end{bmatrix}$$

The **trace** of A is the sum of the diagonal entries of A; that is

$$\mathrm{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn}.$$

Theorem 30.2. *Let A be $n \times n$ matrix with eigenvalues $\lambda_1, \ldots, \lambda_n$. Then*

- $tr(A) = \lambda_1 + \lambda_2 + \cdots + \lambda_n$.
- $\det(A) = \lambda_1 \cdot \lambda_2 \cdot \cdots \cdot \lambda_n$.

Theorem 30.3. *Let A be a 2×2-matrix. Then the characteristic polynomial of A is*

$$p(\lambda) = \lambda^2 - tr(A)\lambda + \det(A).$$

31 Markov matrices

Definition 31.1. An $n \times n$ matrix A is a **Markov matrix** (or: stochastic matrix) if it has only non-negative entries, and the entries in each column add up to 1.
A vector in $\mathbb{R}^n$ is a **probability vector** (or: stochastic vector) if has only non-negative entries, and the entries add up to 1.

Theorem 31.1. *Let A be a Markov matrix. Then*

- *1 is an eigenvalue of A and every other eigenvalue λ of A satisfies $|\lambda| \leq 1$.*
- *If A has only positive entries, then any other eigenvalue satisfies $|\lambda| < 1$.*

Definition 31.2. A **stationary** probability vector of a Markov matrix is probability vector $\mathbf{v}$ that is an eigenvector of A corresponding to the eigenvaule 1.

Theorem 31.2. *Let A be an $n \times n$-Markov matrix with only positive entries and let $\mathbf{z} \in \mathbb{R}^n$ be a probability vector. Then*

$$\mathbf{z}_\infty := \lim_{k \to \infty} A^k \mathbf{z} \text{ exists,}$$

and $\mathbf{z}_\infty$ is a stationary probability vector of A (ie. $A\boldsymbol{z}_\infty = \boldsymbol{z}_\infty$).

32 Diagonalization

Definition 32.1. A square matrix A is said to be **diagonalizable** if there is a invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$.

Theorem 32.1. *Let A be an $n \times n$ matrix that has n linearly independent eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ with associated eigenvalues $\lambda_1, \ldots, \lambda_n$. Then A is diagonalizable as PDP^{-1}, where*

$$P = \begin{bmatrix} \mathbf{v}_1 & \ldots & \mathbf{v_n} \end{bmatrix} \text{ and } D = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$$

Definition 32.2. Vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ form an **eigenbasis** of $n \times n$ matrix A if $\mathbf{v}_1, \ldots, \mathbf{v}_n$ form a basis of $\mathbb{R}^n$ and $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are all eigenvectors of A.

Theorem 32.2. *The following are equivalent for $n \times n$ matrix A:*

- *A has an eigenbasis.*
- *A is diagonalizable.*
- *The geometric multiplicities of all eigenvalues of A sums up to n.*

Theorem 32.3. *Let A be $n \times n$ matrix and let $\mathcal{B} = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be an eigenbasis of A. Then there is a diagonal matrix D such that*

$$A = I_{\mathcal{E}_n \mathcal{B}} D I_{\mathcal{B}\mathcal{E}_n}.$$

33 Powers of Matrices

Key idea: If a matrix A has an eigenbasis, then it is easy to compute A^k, even for large k.

Theorem 33.1. *If $A = PDP^{-1}$, where D is a diagonal matrix, then for any m,*

$$A^m = PD^m P^{-1}.$$

34 Matrix Exponential

Definition 34.1. Let A be an $n \times n$-matrix. We define the **matrix exponential** e^{At} as

$$e^{At} = \sum_{k \geq 0} \frac{(At)^k}{k!} = I + At + \frac{(At)^2}{2!} + \frac{(At)^3}{3!} + \ldots$$

Theorem 34.1. *Let A be an $n \times n$ matrix. Then*

- *the series e^{At} always converges,*
- $e^{At} e^{-At} = I_n$,
- $e^{At} e^{As} = e^{A(t+s)}$,
- $\frac{d}{dt}(e^{At}) = Ae^{At}$.

Theorem 34.2. *Let A be a $n \times n$ matrix such that $A = PDP^{-1}$ for some invertible matrix P and some diagonal matrix D. Then*

$$e^{At} = Pe^{Dt}P^{-1}.$$

Linear Differential Equations

Definition 34.2. A **linear (first order) differential equation** is an equation of the form

$$\frac{d\mathbf{u}}{dt} = A\mathbf{u}$$

where $\mathbf{u}$ is function from $\mathbb{R}$ to $\mathbb{R}^n$. A further condition $\mathbf{u}(0) = \mathbf{v}$, for some $\mathbf{v}$ in $\mathbb{R}^n$ is called a **initial condition**.

Theorem 34.3. *Let A be an $n \times n$ matrix and $\mathbf{v} \in \mathbb{R}^n$. The solution of the differential equation $\frac{d\mathbf{u}}{dt} = A\mathbf{u}$ with initial condition $\mathbf{u}(0) = \mathbf{v}$ is $\mathbf{u}(t) = e^{At}\mathbf{v}$.*

Theorem 34.4. *Let A be an $n \times n$-matrix, and let $\mathbf{v} \in \mathbb{R}^n$ be an eigenvector of A with eigenvalue λ, then $e^{At}\mathbf{v} = e^{\lambda t}\mathbf{v}$.*

Theorem 34.5. *Let A be an $n \times n$-matrix and $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ be an eigenbasis of A with eigenvalues $\lambda_1, \ldots, \lambda_n$. If $\mathbf{v} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$, then the unique solution to the differential equation $\frac{d\mathbf{u}}{dt} = A\mathbf{u}$ with initial condition $\mathbf{u}(0) = \mathbf{v}$ is*

$$e^{At}\mathbf{v} = c_1 e^{\lambda_1 t}\mathbf{v}_1 + \cdots + c_n e^{\lambda_n t}\mathbf{v}_n.$$

35 Orthogonal Projections Onto a Line

Definition 35.1. Let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$. The **orthogonal projection** of $\mathbf{v}$ onto the line spanned by $\mathbf{w}$ is

$$\text{proj}_{\mathbf{w}}\mathbf{v} := \frac{\mathbf{w} \cdot \mathbf{v}}{\mathbf{w} \cdot \mathbf{w}}\mathbf{w}.$$

Theorem 35.1. *Let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$. Then $\text{proj}_{\mathbf{w}}\mathbf{v}$ is the point in $\text{Span}(\mathbf{w})$ closest to $\mathbf{v}$; that is*

$$\text{dist}(\mathbf{v}, \text{proj}_{\mathbf{w}}\mathbf{v}) = \min_{\mathbf{u} \in \text{Span}(\mathbf{w})} \text{dist}(\mathbf{v}, \mathbf{u}).$$

Remark. Note that $\mathbf{v} - \text{proj}_{\mathbf{w}}\mathbf{v}$ (called the **error term**) is in $\text{Span}(\mathbf{w})^{\perp}$.

$$\mathbf{v} = \underbrace{\text{proj}_{\mathbf{w}}\mathbf{v}}_{\in \text{Span}(\mathbf{w})} + \underbrace{\mathbf{v} - \text{proj}_{\mathbf{w}}\mathbf{v}}_{\in \text{Span}(\mathbf{w})^{\perp}}$$

Theorem 35.2. *Let $\mathbf{w} \in \mathbb{R}^n$, then for all $\mathbf{v} \in \mathbb{R}^n$*

$$\text{proj}_{\mathbf{w}}\mathbf{v} = \left(\frac{1}{\mathbf{w} \cdot \mathbf{w}}\mathbf{w}\mathbf{w}^T\right)\mathbf{v}.$$

Remark. Note that $\left(\frac{1}{\mathbf{w} \cdot \mathbf{w}}\mathbf{w}\mathbf{w}^T\right)$ is an $n \times n$ matrix, we call the **orthogonal projection matrix** onto $\text{Span}(\mathbf{w})$.

36 Orthogonal Projection Onto a Subspace

Theorem 36.1. *Let W be a subspace of $\mathbb{R}^n$ and $\mathbf{v} \in \mathbb{R}^n$. Then $\mathbf{v}$ can be written uniquely as*

$$\mathbf{v} = \underbrace{\hat{\mathbf{v}}}_{\text{in } W} + \underbrace{\mathbf{v}^{\perp}}_{\text{in } W^{\perp}}$$

Definition 36.1. We say $\hat{\mathbf{v}}$ is the **orthogonal projection** of $\mathbf{v}$ onto W - written $\text{proj}_W \mathbf{v}$.

Remark. If $(\mathbf{w}_1, \ldots, \mathbf{w}_m)$ is an ***orthogonal basis*** of W, then

$$\text{proj}_W \mathbf{v} = \left(\frac{\mathbf{v} \cdot \mathbf{w}_1}{\mathbf{w}_1 \cdot \mathbf{w}_1}\right) \mathbf{w}_1 + \ldots + \left(\frac{\mathbf{v} \cdot \mathbf{w}_m}{\mathbf{w}_m \cdot \mathbf{w}_m}\right) \mathbf{w}_m.$$

Theorem 36.2. *The projection map $\text{proj}_W : \mathbb{R}^n \to \mathbb{R}^n$ that sends $\mathbf{v}$ to $\text{proj}_W \mathbf{v}$ is linear.*

Definition 36.2. Let matrix P_W be the matrix $(\text{proj}_W)_{\mathcal{E}_n, \mathcal{E}_n}$ that represents proj_W with respect to the standard basis, we call P_W the **orthogonal projection matrix** onto W.

37 Least Squares Solutions

Goal: Suppose $A\mathbf{x} = \mathbf{b}$ is inconsistent. Can we still find something like a *best* solution?

Definition 37.1. Let A be an $m \times n$ matrix and $\mathbf{b} \in \mathbb{R}^m$. A **least squares solution** (short: LSQ solution) of the system $A\mathbf{x} = \mathbf{b}$ is a vector $\hat{\mathbf{x}} \in \mathbb{R}^n$ such that

$$\text{dist}(A\hat{\mathbf{x}}, \mathbf{b}) = \min_{\mathbf{x} \in \mathbb{R}^n} \text{dist}(A\mathbf{x}, \mathbf{b}).$$

Theorem 37.1. *Let A be an $m \times n$ matrix and $\mathbf{b} \in \mathbb{R}^m$. Then $\hat{\mathbf{x}}$ is an LSQ solution to $A\mathbf{x} = \mathbf{b}$ if and only if $A\hat{\mathbf{x}} = \text{proj}_{\text{Col}(A)} \mathbf{b}$.*

Theorem 37.2. *Let A be an $m \times n$ matrix and $\mathbf{b} \in \mathbb{R}^m$. Then $\hat{\mathbf{x}}$ is an LSQ solution to $A\mathbf{x} = \mathbf{b}$ if and only if $A^T A\hat{\mathbf{x}} = A^T \mathbf{b}$.*

Theorem 37.3. *Let A be an $m \times n$ matrix with linearly independent columns and $\mathbf{b} \in \mathbb{R}^m$. Then*

$$\text{proj}_{\text{Col}(A)} \mathbf{b} = A(A^T A)^{-1} A^T \mathbf{b}.$$

38 Linear Regression

There are definitions or theorems in this module, see the lecture notes for examples.

39 Gram-Schmidt Method

Theorem 39.1. *Every subspace of $\mathbb{R}^n$ has an orthonormal basis.*

Algorithm *(Gram-Schmidt orthonormalization)*: Given a basis $\mathbf{a}_1, \ldots, \mathbf{a}_m$, we produce an *orthogonal basis* $\mathbf{b}_1, \ldots, \mathbf{b}_m$ and an *orthonormal basis* $\mathbf{q}_1, \ldots, \mathbf{q}_m$.

$$\begin{array}{ll} \mathbf{b}_1 = \mathbf{a}_1, & \mathbf{q}_1 = \frac{\mathbf{b}_1}{\|\mathbf{b}_1\|} \\ \mathbf{b}_2 = \mathbf{a}_2 - \underbrace{\text{projSpan}(\mathbf{q_1})\mathbf{a}_2}_{=(\mathbf{a}_2\cdot\mathbf{q}_1)\mathbf{q}_1}, & \mathbf{q}_2 = \frac{\mathbf{b}_2}{\|\mathbf{b}_2\|} \\ \mathbf{b}_3 = \mathbf{a}_3 - \underbrace{\text{projSpan}(\mathbf{q}_1, \mathbf{q}_2)\mathbf{a}_3}_{(\mathbf{a}_3\cdot\mathbf{q}_1)\mathbf{q}_1 + (\mathbf{a}_3\cdot\mathbf{q}_2)\mathbf{q}_2} & \mathbf{q}_3 = \frac{\mathbf{b}_3}{\|\mathbf{b}_3\|} \\ \ldots & \ldots \end{array}$$

Remark. Notice that:

- $\text{Span}(\mathbf{q}_1, \ldots, \mathbf{q}_i) = \text{Span}(\mathbf{a}_1, \ldots, \mathbf{a}_i)$ for $i = 1, \ldots, m$, and
- $\mathbf{q}_j \notin \text{Span}(\mathbf{a}_1, \ldots, \mathbf{a}_i)$ for all $j > i$.

Theorem 39.2. (QR decomposition) *Let A be an $m \times n$ matrix of rank n. There is is an $m \times n$-matrix Q with orthonormal columns and an upper triangular $n \times n$ invertible matrix R such that $A = QR$.*

40 The Spectral Theorem

Theorem 40.1. *Let A be a symmetric $n \times n$ matrix, then A has an orthonormal basis of eigenvectors.*

Theorem 40.2. *Let A be a symmetric $n \times n$ matrix. Then there is a diagonal matrix D and a matrix Q with orthonormal columns such that $A = QDQ^T$.*

41 Singular Value Decomposition

Definition 41.1. Let A be an $m \times n$ matrix. A **singular value decomposition** of A is a decomposition $A = U\Sigma V^T$ where

- U is an $m \times m$ matrix with orthonormal columns,
- Σ is an $m \times n$ rectangular diagonal matrix with non-negative numbers on the diagonal,
- V is an $n \times n$ matrix with orthonormal columns.

Remark. The diagonal entries $\sigma_i = \Sigma_{ii}$ which are positive are called the **singular values** of A. We usually arrange them in decreasing order, that is $\sigma_1 \geq \sigma_2 \geq \ldots$

Remark. Let A be an $m \times n$ matrix with rank r. Recall that

- $\text{Nul}(A^TA) = \text{Nul}(A)$ and $\text{Nul}(AA^T) = \text{Nul}(A^T)$.
- A^TA is symmetric and has rank r.

Algorithm 41.1. Let A be an $m \times n$ matrix with rank r.

- Find orthonormal eigenbasis $(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ of A^TA with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_r > \lambda_{r+1} = 0 = \cdots = \lambda_n$.
- Set $\sigma_i = \sqrt{\lambda_i}$ for $i = 1, \ldots, n$.
- Set $\mathbf{u}_1 = \frac{1}{\sigma_1} A\mathbf{v}_1$, ..., $\mathbf{u}_r = \frac{1}{\sigma_r} A\mathbf{v}_r$. (**Magic**: orthonormal!)
- Find $\mathbf{u}_{r+1}, \ldots, \mathbf{u}_m \in \mathbb{R}^m$ such that $(\mathbf{u}_1, \ldots, \mathbf{u}_m)$ is an orthonormal basis of $\mathbb{R}^m$.
- Set

$$U = \begin{bmatrix} \mathbf{u}_1 & \ldots & \mathbf{u}_m \end{bmatrix}, \Sigma = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_{\min\{m,n\}} \end{bmatrix}, V = \begin{bmatrix} \mathbf{v}_1 & \ldots & \mathbf{v}_n \end{bmatrix}$$

Notice that $\mathbf{v}_{r+1}, \ldots, \mathbf{v}_n \in \text{Nul}(A^TA)$, and hence $\mathbf{v}_{r+1}, \ldots, \mathbf{v}_n \in \text{Nul}(A)$.

$$\begin{aligned} \rightsquigarrow AV &= A \begin{bmatrix} \mathbf{v}_1 & \ldots & \mathbf{v}_n \end{bmatrix} = \begin{bmatrix} A\mathbf{v}_1 & \ldots & A\mathbf{v}_n \end{bmatrix} \\ &= \begin{bmatrix} \sigma_1\mathbf{u}_1 & \ldots & \sigma_r\mathbf{u}_r & \mathbf{0} & \ldots & \mathbf{0} \end{bmatrix} = U\Sigma \rightsquigarrow A = U\Sigma V^T. \end{aligned}$$

Theorem 41.1. *Let A be an $m \times n$ matrix with rank r, and suppose $U = \begin{bmatrix} \mathbf{u}_1 & \ldots \mathbf{u}_m \end{bmatrix}$, $V = \begin{bmatrix} \mathbf{v}_1 & \ldots \mathbf{v}_n \end{bmatrix}$, and Σ are such that $A = U\Sigma V^T$ is a SVD of A. Then*

- $(\mathbf{u}_1, \ldots, \mathbf{u}_r)$ *is a basis of* $\text{Col}(A)$.
- $(\mathbf{u}_{r+1}, \ldots, \mathbf{u}_m)$ *is a basis of* $\text{Nul}(A^T)$.
- $(\mathbf{v}_1, \ldots, \mathbf{v}_r)$ *is a basis of* $\text{Col}(A^T)$.
- $(\mathbf{v}_{r+1}, \ldots, \mathbf{v}_n)$ *is a basis of* $\text{Nul}(A)$.

42 Low Rank Approximation

Theorem 42.1. *Let A be an $m \times n$ matrix with rank r, and let $U = \begin{bmatrix} \mathbf{u}_1 & \ldots \mathbf{u}_m \end{bmatrix}$, $V = \begin{bmatrix} \mathbf{v}_1 & \ldots \mathbf{v}_n \end{bmatrix}$ be matrices with with orthonormal columns and Σ be a rectangular diagonal $m \times n$ matrix such that $A = U\Sigma V^T$ is an SVD of A. Then*

$$\begin{aligned} A &= \sigma_1\mathbf{u}_1\mathbf{v}_1^T + \sigma_2\mathbf{u}_2\mathbf{v}_2^T + \cdots + \sigma_r\mathbf{u}_r\mathbf{v}_r^T \\ &= \begin{bmatrix} \mathbf{u}_1 & \ldots & \mathbf{u}_r \end{bmatrix} \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix} \begin{bmatrix} \mathbf{v}_1 & \ldots & \mathbf{v}_r \end{bmatrix}^T. \end{aligned}$$

Definition 42.1. For $k \leq r$, define

$$A_k = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \cdots + \sigma_k \mathbf{u}_k \mathbf{v}_k^T .$$

Idea. If $\sigma_1 \gg \sigma_2 \gg \ldots ...$, then A_k is a good approximation of A.

Definition 42.2. Let A be an $m \times n$ matrix with rank r. A **compact singular value decomposition** of A is a decomposition $A = U_c \Sigma_c V_c^T$ where

- $U_c = \begin{bmatrix} \mathbf{u}_1 & \ldots \mathbf{u}_r \end{bmatrix}$ is an $m \times r$ matrix with orthonormal columns,
- Σ_c is an $r \times r$ diagonal matrix with positive diagonal elements,
- $V_c = \begin{bmatrix} \mathbf{v}_1 & \ldots \mathbf{v}_r \end{bmatrix}$ is an $n \times r$ matrix with orthonormal columns.

43 Pseudoinverses

Definition 43.1. Let A be an $m \times n$ matrix with rank r. Given the compact singular value decomposition $A = U_c \Sigma_c V_c^T$, where

- $U_c = \begin{bmatrix} \mathbf{u}_1 & \ldots \mathbf{u}_r \end{bmatrix}$ is an $m \times r$ matrix with orthonormal columns,
- Σ_c is an $r \times r$ diagonal matrix with positive diagonal elements,
- $V_c = \begin{bmatrix} \mathbf{v}_1 & \ldots \mathbf{v}_r \end{bmatrix}$ is an $n \times r$ matrix with orthonormal columns,

we define the **pseudoinverse** A^+ of A as $V_c \Sigma_c^{-1} U_c^T$.

Theorem 43.1. *Let* $\mathbf{v} \in \mathrm{Col}(A^T)$ *and* $\mathbf{w} \in \mathrm{Col}(A)$. *Then* $A^+ A\mathbf{v} = \mathbf{v}$ *and* $AA^+\mathbf{w} = \mathbf{w}$.

Remark. If A is $n \times n$ and invertible, then $\mathrm{Col}(A) = \mathbb{R}^n$. Thus $A^{-1} = A^+$.

Theorem 43.2. *Let* A *be an* $m \times n$ *matrix and let* $\mathbf{b} \in \mathbb{R}^m$. *Then* $A^+\mathbf{b}$ *is the LSQ solution of* $A\mathbf{x} = \mathbf{b}$ *(with minimum length).*

Remark. This is particularly useful, when solving many different LSQ problems of the form $A\mathbf{x} = \mathbf{b}$, where A stays the same, but $\mathbf{b}$ varies.

44 Principal Component Analysis

Setup:

- Given m objects, we measure the same n variables.
- Thus m samples of n-dimensional data $\rightsquigarrow$ $m \times n$ matrix (each row is a sample)
- Analyse this matrix to understand what drives the variance in the data.

Definition 44.1. Let $X = \begin{bmatrix} \mathbf{a}_1 & \dots & \mathbf{a}_m \end{bmatrix}^T$ be an $m \times n$ matrix. We define the **column average** $\mu(X)$ of X as

$$\mu(X) := \frac{1}{m}(\mathbf{a}_1 + \cdots + \mathbf{a}_m).$$

We say X is **centered** if $\mu(X) = \mathbf{0}$, and for X centered, we define the **covariance matrix** $\operatorname{cov}(X)$ of X as $\frac{1}{m-1}X^TX$.

Principal component analysis (PCA):

- Input: centered $m \times n$-matrix X.
- Compute $\operatorname{cov}(X)$.
- Since $\operatorname{cov}(X)$ is symmetric, we can find an orthonormal eigenbasis $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ of $\operatorname{cov}(X)$ with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$.
- Write $\operatorname{cov}(X)$ as a sum of rank 1 matrices:

$$\operatorname{cov}(X) = \lambda_1 \mathbf{v}_1 \mathbf{v}_1^T + \cdots + \lambda_n \mathbf{v}_n \mathbf{v}_n^T.$$

- Each principal component $\mathbf{v}_i$ explains part of the variance of the data. The larger λ_i, the more of the variances is explained by $\mathbf{v}_i$.

Remark. PCA using SVD.

- Let X be a centered data matrix. Observe that $X^TX = (m-1)\operatorname{cov}(X)$.
- To find orthonormal eigenbasis of $\operatorname{cov}(X)$, it is enough to find orthonormal eigenbasis of X^TX.
- Compute the SVD of $X \rightsquigarrow X = U\Sigma V^T$.
- The columns of $V = \begin{bmatrix} \mathbf{v}_1 & \dots & \mathbf{v}_n \end{bmatrix}$ are the desired orthonormal eigenbasis.
- If σ_i is the singular value for $\mathbf{v_i}$, then

$$\lambda_i = \frac{\sigma_i^2}{m-1}$$

45 Review of Complex Numbers

Definition 45.1. $\mathbb{C} = \{x + iy \mid x, y \in \mathbb{R}\}$ where $i = \sqrt{-1}$, or $i^2 = -1$.

- The **real part** of z, denoted $\Re(z)$ is defined by $\Re(z) = x$.
- The **imaginary part** of z, denoted $\Im(z)$ is defined by $\Im(z) = y$.
- The **complex conjugate** of z, denoted $\overline{z}$, is defined by $\overline{z} = x - iy$.

- The **absolute value**, or **magnitude** of z, denoted $|z|$ or $\|z\|$, is given by

$$|z| = \sqrt{x^2 + y^2}.$$

Remark. Any point in $\mathbb{R}^2$ can be viewed as a complex number: $\binom{x}{y} \leftrightarrow x + iy$.

Definition 45.2. Given $z = x+iy$, $w = u+iv$, we define $z+w = (x+u)+i(y+v)$, and

$$\begin{aligned} zw &= (x + iy)(u + iv) \\ &= xu + x(iv) + (iy)u + (iy)(iv) \\ &= (xu - yv) + i(xv + yu) \end{aligned}$$

Theorem 45.1. *Let $z \in \mathbb{C}$.*

- $\overline{\overline{z}} = z$
- $|z|^2 = z\overline{z}$
- $|z| = |\overline{z}|$

46 Complex Linear Algebra

Goal. Use complex numbers (instead of real numbers) as scalars.

Definition 46.1. The (complex) vector space $\mathbb{C}^n$ is of all *complex* column vectors $\mathbf{z} = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix}$, where $z_1, z_2, \ldots, z_n$ are complex numbers.

Remark.

- Now multiplication by a *complex scalar* makes sense.
- We can define subspaces, Span, independence, basis, dimension for $\mathbb{C}^n$ in the usual way.
- We can multiply complex vectors by complex matrices. Column space and Null space still make sense.
- The only difference is the dot product, you need to use the complex conjugate to get a good notion of length:

$$\begin{bmatrix} z_1 \\ \vdots \\ z_n \end{bmatrix} \cdot \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix} = z_1\overline{w}_1 + z_2\overline{w}_2 + \ldots z_n\overline{w}_n.$$

Definition 46.2. Let A be an $m \times n$-matrix. The **conjugate matrix** $\overline{A}$ of A is obtained from A by taking the complex conjugate of each entry of A.

Theorem 46.1. *Let A be a matrix with real entries and λ is an eigenvalue of A. Then $\bar{\lambda}$ is also a eigenvalue. Furthermore, if $\mathbf{v}$ is an eigenvector with eigenvalue λ, then $\bar{\mathbf{v}}$ is an eigenvector with eigenvalue $\bar{\lambda}$.*

Definition 46.3. Let A be an $m \times n$-matrix. The **conjugate transpose** A^H of A is defined as $\overline{A}^T$. We say the matrix A is **Hermitian** if $A = A^H$.